

November 13, 2025

Constants used

$$\begin{aligned}
h &= 6.62607015 \cdot 10^{-34} \text{ J} \cdot \text{s} \\
c &= 2.99792458 \cdot 10^8 \text{ m/s} \\
e &= 1.602176634 \cdot 10^{-19} \text{ C} \\
m_e &= 9.11 \cdot 10^{-31} \text{ kg}
\end{aligned}$$

Problems and solutions

1. A light source has a power output of 50 W.

- (a) What is the frequency of the light if the entire energy of the source is emitted as light of wavelength $\lambda = 500 \text{ nm}$?

The frequency is

$$f = \frac{c}{\lambda} = \frac{2.99792458 \cdot 10^8}{5 \cdot 10^{-7}} = 5.995849 \cdot 10^{14} \text{ Hz}.$$

- (b) How many photons are emitted by the light source per second?

Photon energy $E_{ph} = hf$. The photon emission rate is $\frac{\Delta N}{\Delta t} = P/E_{ph}$:

$$E_{ph} = hf = 3.972892 \cdot 10^{-19} \text{ J}, \quad \frac{\Delta N}{\Delta t} = \frac{50}{E_{ph}} = 1.258529 \cdot 10^{20} \text{ s}^{-1}.$$

2. How many photons per second are emitted by a 5 mW carbon dioxide laser if its wavelength is $10 \mu\text{m}$?

Photon energy $E_{ph} = hc/\lambda$. With $P = 5 \cdot 10^{-3} \text{ W}$,

$$E_{ph} = 1.986446 \cdot 10^{-20} \text{ J}, \quad \frac{\Delta N}{\Delta t} = \frac{5 \cdot 10^{-3}}{E_{ph}} = 2.517058 \cdot 10^{17} \text{ s}^{-1}.$$

3. A laser pulse of energy $E = 16 \text{ J}$ has a duration $\tau = 4 \cdot 10^{-7} \text{ s}$. The light, of wavelength $\lambda = 600 \text{ nm}$, strikes a metal surface.

- (a) What is the power of the laser?

$$P = E/\tau = 4.00 \cdot 10^7 \text{ W}.$$

- (b) How many photons strike the metal surface?

Photon energy $E_{ph} = hc/\lambda = 3.310743 \cdot 10^{-19} \text{ J}$. Number of photons in the pulse $N = E/E_{ph} = 4.832752 \cdot 10^{19}$.

- (c) What is the maximum velocity of the electrons ejected from the metal by the light, if the work function is $4 \cdot 10^{-19} \text{ J}$? (Electron mass: $m_e = 9.11 \cdot 10^{-31} \text{ kg}$)

Maximum kinetic energy $E_{kin}^{\max} = E_{ph} - \phi = -6.892569 \cdot 10^{-20} \text{ J}$. Therefore, no electrons are ejected from the metal by the light with these parameters.

4. What is the energy of a photon of X-ray radiation with a wavelength of $5 \cdot 10^{-11} \text{ m}$, in joules and in electronvolts? What are the momentum and mass of the photon?

$$E_{ph} = \frac{hc}{\lambda} = 3.972892 \cdot 10^{-15} \text{ J} = 2.479684 \cdot 10^4 \text{ eV},$$

$$p = \frac{E_{ph}}{c} = 1.325214 \cdot 10^{-23} \text{ kg} \cdot \text{m/s}, \quad m_{eff} = \frac{E_{ph}}{c^2} = 4.420438 \cdot 10^{-32} \text{ kg}.$$

5. A metal surface is illuminated with light of wavelength $\lambda_1 = 492 \text{ nm}$. The maximum kinetic energy of the emitted electrons is $W_{kin1} = 9.9 \cdot 10^{-19} \text{ J}$. When the wavelength of the light is changed to $\lambda_2 = 579 \text{ nm}$, the kinetic energy decreases to $W_{kin2} = 3.8 \cdot 10^{-20} \text{ J}$. Based on these data, determine the Planck constant and the work function characteristic of the metal!

From $E_{kin}^i = \frac{hc}{\lambda_i} - \phi$ (for $i = 1, 2$) subtracting yields

$$h = \frac{E_{kin}^1 - E_{kin}^2}{c \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)} = 1.039778 \cdot 10^{-32} \text{ J} \cdot \text{s}.$$

Then $\phi = \frac{hc}{\lambda_1} - E_{kin}^1 = 5.345724 \cdot 10^{-18} \text{ J}$.

6. When light of a certain frequency is incident on a metal surface, the measured stopping potential of the emitted electrons is $U_{f1} = 3.49 \text{ V}$. For light of half that frequency, the stopping potential is $U_{f2} = 0.75 \text{ V}$. Determine the work function and the wavelength of the light!

Using $eU_s = hf - \phi$ for frequency f and $f/2$, subtracting gives $hf = 2e(U_{s1} - U_{s2})$. Hence

$$f = \frac{2e(U_{s1} - U_{s2})}{h} = 1.325058 \cdot 10^{15} \text{ Hz}, \quad \lambda = \frac{c}{f} = 2.262485 \cdot 10^{-7} \text{ m}.$$

Work function:

$$\phi = hf - eU_{s1} = 3.188332 \cdot 10^{-19} \text{ J} = 1.99 \text{ eV}.$$

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